GENERAL SIR JOHN KOTELAWALA DEFENCE UNIVERSITY
FACULTY OF COMPUTING
DEPARTMENT OF INFORMATION TECHNOLOGY
INTAKE 41



ESSENTIALS OF ARTIFICIAL INTELLIGENCE
PROJECT PROPOSAL

Group Details			
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Project Title	AI-Based Herbal Plant Identification and Digital Herbarium System - CeylonHerb		

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1. Introduction

A wide range of herbal plants are utilized in traditional medicine, home cures, and education in Sri Lanka. However, because many plants have similar visual characteristics, precisely recognizing them can be challenging.

CeylonHerb is a digital herbarium and AI-based system for identifying herbal plants. To obtain plant predictions, confidence scores, and botanical data, users can upload or take a picture of a leaf. The system will identify ten to fifteen specific Sri Lankan herbal plants using **CNN-based classification, rule-based reasoning, and similarity search.**

2. Background of the Selected Problem

In Sri Lanka, identifying herbal plants frequently relies on literature, online, or professional expertise. When similar-looking plants are involved, these methods can be inconsistent, slow, and difficult to access.

By analyzing leaf characteristics including form, color, texture, and vein patterns, AI-based computer vision can enhance this procedure. But a basic picture classifier might merely yield the name of the plant. As a result, **CeylonHerb** offers forecasts, verification, recommendations for comparable plants, and organized educational content by combining AI-based identification with a digital herbarium.

3. Problem Statement

Due to a **lack of botanical expertise, similar-looking species, untrustworthy internet material, and restricted access to experts**, Sri Lankan herbal plants may be difficult for researchers, students, home gardeners, and the general public to correctly identify. Additionally, incorrect identification may pose a risk to one's health and safety.

As a result, **CeylonHerb** will evaluate leaf photos, forecast potential herbal plants, offer confidence ratings, confirm observable characteristics, and present trustworthy botanical data. The approach is not meant for medical diagnosis or treatment recommendations, but rather for **plant identification and education**.

4. Objectives of the Proposed AI Solution

Main Objective

To develop an AI-based web application that uses photos of leaves to identify certain Sri Lankan herbal plants and offers relevant data via a digital herbarium.

Specific Objectives

1. Create a labeled dataset containing approximately 10 - 15 herbal plant species.
2. Develop a CNN-based model for classifying leaf images.
3. Display the confidence scores for the top three predictions.
4. Verify uncertain predictions using rule-based reasoning.
5. Create a digital herbarium that can be searched.
6. Allow plant searches using local, English, scientific, or botanical family names.
7. Suggest visually similar plants.
8. Provide warnings for low-confidence predictions.
9. Test the model using unseen images.
10. Evaluate the model using accuracy, precision, recall, F1-score, and a confusion matrix.

5. Target Users

Primary Users

- University students
- School students
- Home gardeners
- Herbal plant enthusiasts
- Botany beginners
- Ayurveda and indigenous medicine students

Secondary Users

- Lecturers and teachers
- Researchers
- Plant nursery workers
- Agricultural officers
- Environmental organisations
- Herbal garden maintainers

6. Proposed AI Techniques

CNN-Based Image Classification

A **CNN** will identify herbal plants by learning leaf features such as **shape, edges, colour, texture, and vein patterns**. The project will use **transfer learning** with a pretrained model such as MobileNetV2 or EfficientNetB0 to generate plant prediction probabilities.

Rule-Based Expert System

If the CNN confidence is low or medium, the system will ask about visible features such as **leaf shape, edge, arrangement, and plant type**. Simple **IF–THEN rules** will then help confirm or adjust the prediction.

Example: **IF** leaf shape is oval, leaf edge is serrated, and arrangement is opposite, **THEN** suggest Plant A.

Content-Based Similarity Search

The system will compare image features using methods such as **Cosine Similarity, Euclidean Distance, or K-Nearest Neighbours** to find visually similar plants. This helps users compare possible species instead of depending on a single prediction.

7. Justification for Selecting the AI Techniques

CNN and Transfer Learning: Ideal for examining leaf images and identifying characteristics including edges, texture, colour, and form. Transfer learning is effective with smaller datasets and cuts down on training time.

Rule-Based Reasoning: Helps verify uncertain CNN predictions using simple **IF–THEN rules**, especially when plants look similar.

Similarity Search: Helps users compare the uploaded leaf with visually similar plant images in the herbarium.

Therefore, the project demonstrates three AI concepts: **deep learning, knowledge-based reasoning, and intelligent search/recommendation**.

8. Expected System Inputs and Outputs

Inputs: The system will accept a leaf image, search text, basic plant features such as shape and leaf edge, user feedback, and stored herbarium information.

Outputs: The system will provide the predicted plant, top possible matches, a confidence score, a rule-based verification result, relevant herbarium information, similar plants, warning messages for uncertain predictions, and basic model evaluation results.

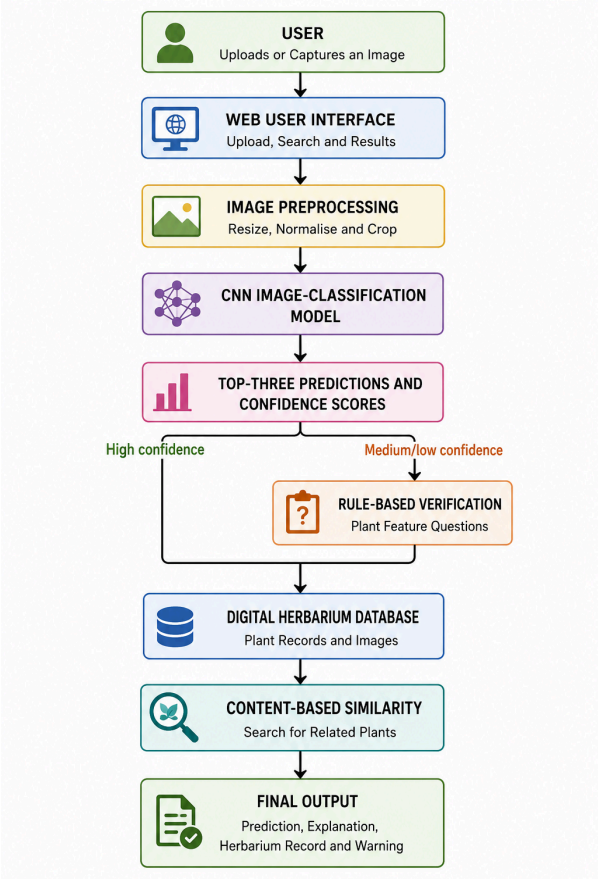
Example Output
For instance, the system might identify a photograph uploaded by the user as Gotu Kola with an 87% confidence level. Other potential plants, including water clover (8% confidence) and pennywort (5% confidence), may also be shown. Before the plant is used medicinally, the system will also show a warning suggesting expert confirmation because the forecast is produced by an experimental AI model.

9. Proposed Tools and Technologies

Purpose	Technology
Programming	Python
AI Model Development	TensorFlow / Keras
Image Processing	OpenCV
Development Environment	Google Colab
Backend	Flask
Frontend	HTML, CSS, JavaScript
Database	MySQL

10. Initial System Diagram / Workflow

10.1 System Architecture



10.2 Workflow

1. The user uploads or captures a leaf image.
2. The system validates and preprocesses the image.
3. The CNN model analyses the image.
4. Top-three predictions and confidence scores are generated.
5. If confidence is low or medium, the rule-based system asks plant-feature questions.
6. The prediction is verified or adjusted.
7. Related plant information is retrieved from the digital herbarium.
8. Similar plants are displayed.
9. The final result, confidence score, plant information, and warning are shown.

11. Work Division Among Members

Member	Responsibility	Main Tasks
WGISK Wedage	Coordination & Research	Research, requirements, timeline, report and submissions
KMGPD Kariyawasam	Dataset & AI Model	Image collection, preprocessing, CNN training and evaluation
PV Kavinda	Backend & AI Reasoning	Flask backend, database, rule-based system and similarity search
KHTU Jayasekara	Frontend & Testing	UI development, testing, documentation, presentation and video

12. Timeline for Completing the Work

	Main Activity	Week									
		1	2	3	4	5	6	7	8	9	10
1	**Confirm problem, scope, users,** and member responsibilities										
2	**Conduct background research,** select plant classes, define objectives and AI techniques										
3	**Design architecture, workflow,** inputs/outputs, tools, and proposal										
4	**Finalise proposal and** begin image collection										
5	**Complete dataset collection,** cleaning, preprocessing, and augmentation										
6	**Train initial CNN model** and begin Flask/database development										
7	**Improve model, create rules,** and develop similarity search										
8	**Integrate AI, database, and** interface; prepare screenshots										
9	**Complete integration and** conduct testing										
10	**Finalise report, presentation, video, source code,** and contribution sheet										

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